Mind Wandering Detection of Meditators Using EEG

Max Boyington

Github | Dataset

Project Purpose

Mind-wandering, also known as daydreaming, or more strictly defined as task-unrelated thought (Smallwood & Schooler, 2006) is a phenomenon when attention drifts away from a task towards an internally generated thought. Mind-wandering occupies a substantial portion of waking life–there have even been claims that people spend 30-50% of their waking lives mind wandering. Importantly, mind-wandering has also been associated with reduced well-being and increased unhappiness (Killingsworth & Gilbert, 2010).

Meditation has been shown to be a promising exercise to reduce mind-wandering, with prior work demonstrating both behavioral and neural differences between advanced and novice meditators (Brandmeyer & Delorme, 2018). This project aims to develop an EEG-based approach to detecting mind-wandering in meditators. The long-term motivation is to lay the groundwork for future neurofeedback systems that could provide real-time, “assisted” meditation support for novice meditation practitioners.

While prior work has demonstrated that EEG-based classifiers can detect mind-wandering and has examined generalization across tasks or contexts, less attention has

been paid to how models can calibrate to individual subjects. This project therefore will focus on comparing generalization to unseen versus partially seen subjects. In addition, this project will evaluate the predictive contribution of different EEG frequency bands as input features, with the goal of replicating results of band-specific effects (Jin et al., 2019; Tang et al., 2023).

Broader Implications

Beyond the immediate goal of a neurofeedback system to “assist” meditation, the development of robust, individually calibrated mind-wandering detection has significant implications for high-stakes environments and neurofeedback technology. In contexts such as long-distance driving or air traffic control, where sustained attention is critical for safety, a mind-wandering detection system could serve as an early-warning system to prevent accidents caused by poor focus. Furthermore, in the modern workplace, such systems could be used to help users to identify peak focus periods or provide interventions to prevent long periods of daydreaming.

Perhaps most importantly, this project aims to explore the importance of individually calibrated models. By comparing the partially-seen subject classifier to the LOSO (leave-one-subject-out) model, we can more clearly see how impactful calibration is to train a classifier. In the real world, a BCI could possibly need the user to give a “calibration period”, where they allow the model to collect 5-10 minutes of resting data, or n number of epochs associated with a brain state (like mind-wandering).

Ultimately, this work aims to demonstrate the importance of neural signal isolation and prioritizing subject-specific calibration for the transition of brain-state monitoring from controlled laboratory environments to reliable, real-world use.

Methods

Dataset

This project used a publicly available EEG dataset from a study that investigated neural correlates of mind-wandering during meditation (Brandmeyer & Delorme, 2018; Delorme & Brandmeyer, 2024). The study included 24 meditators in the Himalayan Yoga tradition, divided in expert and non-expert groups based on meditation experience. Three subjects were excluded due to extreme class imbalance, resulting in 21 subjects being used. The criteria for non-experts were still experienced practitioners, who meditated an average of 3.2 hours per week, while expert meditators were those who engaged in a minimum of 2 hours of daily practice for a year or longer.

EEG data of the subjects meditating was recorded using a 64-channel Biosemi system. The experimental sessions lasted between 45 and 90 minutes, with participants receiving experience-sampling probes at random intervals ranging from 30 to 90 seconds. At each probe, participants used a 0–3 scale to report their meditation depth, mind-wandering depth, and level of fatigue. This project follows the precedent set by the study to use 10 second epochs preceding each probe for the classifier (-10.05 to -0.05 seconds prior).

The epochs were separated into two categories: mind-wandering trials and meditation trials. Since participants likely were to rate questions differently (recall it is a subjective score of 0-3 of how much one is meditating and mind-wandering), some subjects were more biased than others towards rating things higher or lower. Instead of trying to normalize this noise across subjects, I will follow the strategy used in the dataset's paper: trials where ratings on the

meditation scale were larger than mind-wandering would be considered meditation trials, and vice versa for mind-wandering trials. Lastly, trials where the ratings for both conditions were equal were ignored.

After excluding subjects and epochs, the meditating class had 466 samples and the mind-wandering class had 296 samples across 21 subjects.

Preprocessing

The EEG data was already downsampled from 2048 Hz to 256 Hz in the public dataset. While ICA is a common artifact-rejection tool, it was not utilized in this study to adhere to the scope and time constraints of the project. Instead, a minimal preprocessing pipeline was adopted, utilizing a 2 Hz high-pass filter to mitigate low-frequency artifacts while preserving the original signal's characteristics. A low-pass filter at 40 Hz was applied as well. Bad electrodes were removed using the NoisyChannels class in the PyPREP library and then interpolated using MNE. Due to excessive noise in the mastoid channels, the data was re-referenced to the common average of the electrodes.

Feature Extraction

Following the preprocessing pipeline, EEG signals were split into 10-second epochs preceding each probe (-10.05 to -0.05 seconds prior). Then, the power spectral density (PSD) was computed for each of the 64 electrodes using a Fast Fourier Transformation (FFT).

Periodic (Oscillatory) Parameterization Features

To evaluate which spectral representations were the most effective features, three distinct categories of power features across five frequency bands were extracted: Delta (2-4 Hz), Theta (4-8 Hz), Alpha (8-12 Hz), Beta (12-30 Hz), and Gamma (30-40 Hz). For each frequency band, three types of mean band-power values were extracted:

- Absolute Mean Band Power: Represents the raw average power within each frequency band.
- Relative Mean Band Power: The ratio of power in a specific band relative to the total power across the entire 2 - 40Hz spectrum. This was included as relative mean band power can minimize the impact of inter-subject variance.
- FOOOF-Flattened Mean Band Power: The Fitting Oscillations and One-Over-F (FOOOF) algorithm (Donoghue et al., 2020) was employed to separate periodic oscillations from the aperiodic 1/f background. By subtracting the aperiodic component from the PSD, the “flattened” power spectra was calculated, and then separated into the mean band power for each of the 5 frequency bands. This is another method that can reduce the impact of inter-subject variance.

Aperiodic (Background) Parameterization Features

Aperiodic background features were also used as features in the model, specifically the offset and exponent. Unlike traditional spectral analysis that treats the 1/f signal as background noise, recent evidence suggests that the aperiodic slope reflects critical neurophysiological markers, including the balance of excitation and inhibition. Furthermore, the exponent has shown potential

in distinguishing expert meditators from non-experts, suggesting its relevance as a feature for detecting deep meditative states versus mind-wandering (Rodriguez-Larios et al., 2021).

The aperiodic component (L) is modeled using a Lorentzian function:

$$L=b-\log(k+F^{\chi})$$

where *b* represents the offset (y-intercept), χ represents the aperiodic exponent (negative slope), and *k* represents the "knee" parameter. The offset (*b*) and the aperiodic exponent (χ) were extracted as independent features for the classifier and compared with the other features.

The Final Features

The final feature vector for each 10-second epoch was constructed by concatenating the multi-tiered spectral and aperiodic parameters across all 64 electrodes. This high-dimensional input space consisted of 960 periodic features (64 electrodes x 3 mean power types x 5 frequency bands) and 128 aperiodic features (64 electrodes x 2 aperiodic components), resulting in a total of 1,088 features per sample.

Classifiers were trained and evaluated on individual feature categories (such as specific frequency bands like Alpha or the standalone aperiodic set) and compared against groupings, including all mean power, all relative power, all flattened power, both aperiodic components, and all combined features.

Feature Category	Parameters	Description
Mean Band Power	Delta, Theta, Alpha, Beta, Gamma	Raw average power within frequency bins

Relative Mean Band Power	Delta, Theta, Alpha, Beta, Gamma	Band power normalized by total 2-40 Hz power
FOOOF-Flattened Power	Delta, Theta, Alpha, Beta, Gamma	Power after subtracting aperiodic 1/f component
Aperiodic Components	Exponent (χ), Offset (b)	Slope and intercept of the power spectrum

Classifier

Two classifiers were used for this project: a support vector machine (SVM) and XGBoost. While the primary focus was on the SVM, the XGBoost classifier was also evaluated for its ability to handle non-linear relationships in tabular data.

Hyperparameters and Model Pipeline

The SVM with an RBF kernel (radial basis function) was selected instead of a linear kernel due to the high amount of noise in EEG (Jin et al., 2019). Another motivation for both the SVM and RBF kernel is that in other mind-wandering detection studies, it performed the best (Tang et al., 2023).

To mitigate the risk of overfitting the relatively small sample size and large feature input (64 to 1,088 features depending on the feature category), a standardized pipeline was implemented for both models:

- Scaling: A StandardScaler (from sklearn.preprocessing) was used to normalize features ensuring that values across different frequency bands and electrodes were on a comparable scale
- Dimensionality Reduction: Principal Component Analysis (PCA) was applied to reduce the feature space to 10 components, ensuring the model focused only on the most significant data.
- SVM Configuration: The model used balanced class weights to account for the disparity between meditation trials (n=466) and mind-wandering trials (n=296).
- XGBoost Configuration: The model was configured with 100 estimators with a maximum depth of 2 to maintain model simplicity and prevent model overfitting.

Evaluation Techniques

Subject-Level Cross-Validation

To evaluate how well models calibrate to new individuals, two subject-level validation strategies will be implemented:

1. Unseen Subjects (Leave-One-Subject-Out / LOSO Cross Validation): This strategy tested the model's ability to generalize to completely unseen subjects, simulating a "plug-and-play" BCI system with zero calibration.

2. Partially-seen Subjects (Leave out partial subjects): Similar to LOSO, except two subjects will leave out 50% of their data points, giving the training model 50% of the subjects' sessions. This simulates a system that utilizes a personal calibration period, assessing how important subject-specific tuning is for classification accuracy.

Evaluation Metrics and Class Imbalance

To account for the inherent class imbalance in the dataset (466 meditation samples vs. 296 mind-wandering samples), model performance was primarily evaluated using the F1-Macro score. Unlike standard accuracy, the F1-Macro score provides an unweighted average of the precision and recall for each class, ensuring that the model is penalized if it fails to accurately predict the minority (mind-wandering) class. Additionally, balanced class weights were applied within the SVM configuration to further adjust the cost function during training, preventing the classifier from developing a bias toward the more frequent meditation trials.

Label Shuffling

A critical challenge in EEG-based state detection is the risk of the model "memorizing" the unique neural signatures—or fingerprints—of individual participants rather than learning the generalized neural correlates of a brain state. To audit the models for this subject-recognition bias, we implemented a Shuffled Label Baseline.

In this control condition, the target labels (Meditation vs. Mind-Wandering) were randomized within each subject's data before training. If a classifier's performance on the original data is equivalent to its performance on shuffled data, it indicates that the model is utilizing stable, subject-specific spectral traits to "identify" the individual rather than the transitory cognitive state. This technique serves as a necessary benchmark to distinguish between a functional brain-state classifier and an effective subject classifier.

Feature Comparison

To determine what components are most relevant for classification, the performance of each individual feature and feature groups (eg. all mean power, all periodic, all feature combined, etc) across the SVM and XGBoost models were compared. Performance was measured for both original and shuffled labels to observe the classifiers subject-recognition bias. Because initial evaluations revealed that the models were primarily detecting unique individual "fingerprints" of the subjects rather than brain states, the final comparison focus is placed on the shuffled-label scores. This methodological choice was intended to identify which EEG components—such as specific frequency bands or aperiodic parameters—contribute most significantly to stable, subject-specific identification.

Results

Evidence of Subject-Recognition Bias via Shuffled Labels

The primary finding of this study is that the classification models were effectively identifying individual subjects rather than detecting transient cognitive states (Mind-Wandering vs. Meditation). As shown in Figure 1, a "Shuffled Label Baseline" was established by randomizing target labels within each subject's data.

In the Leave2Out (Subject-Independent) condition, the model achieved a mean accuracy of 0.53, which was marginally outperformed by the shuffled baseline of 0.54. Similarly, in the LeavePartialGroupOut (Subject-Calibrated) condition, both the original and shuffled models yielded identical accuracies of 0.61. The fact that the models performed equally well when labels

were randomized confirms that the classifiers were leveraging stable, subject-specific biometric signatures (or "neural fingerprints") rather than the underlying brain state.

Impact of Subject Calibration on Model Performance

The performance improved when moving from completely unseen to partially unseen subject test data. This suggests that a “calibration period” allows the model to map the unique “fingerprint” of a user. However, since the scores of the shuffled baseline rose the same as the unshuffled data, it is likely this calibration is optimized for subject identification and not brain state decoding.

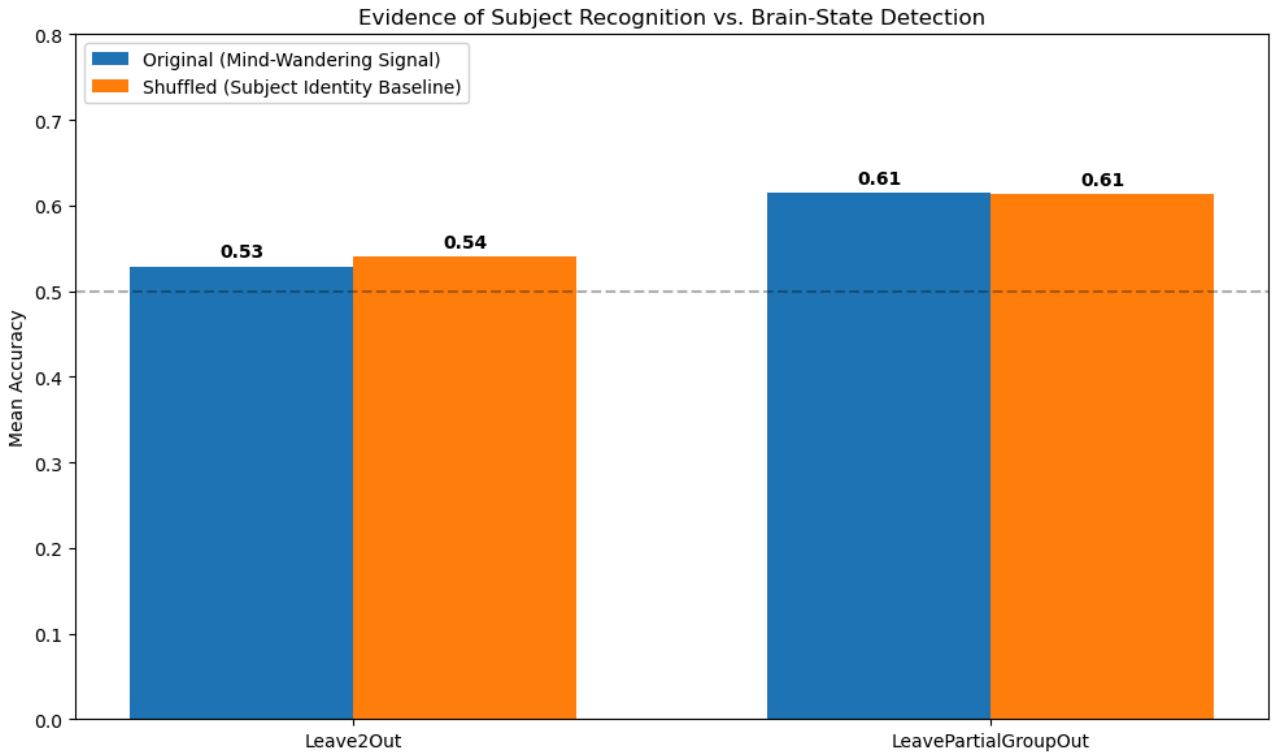

Figure 1: Accuracy comparison between original and shuffled labels across validation strategies. The identical performance in the calibrated group (0.61) confirms the model is predicting subject identity rather than brain states.

Feature Efficacy for Subject Identification

To identify what EEG components were most responsible for success in this model, the top 10 feature-model combinations were ranked based on their performance under the shuffled labels. It can be inferred that these features are most relevant for subject recognition. The highest features on the F1-Macro score were the combined absolute mean power (0.577), combined relative mean power (0.565), and all combined features (0.563), and all were on the SVM. It is likely that the more data the model has to work with, the more capable it is of recognizing the subject.

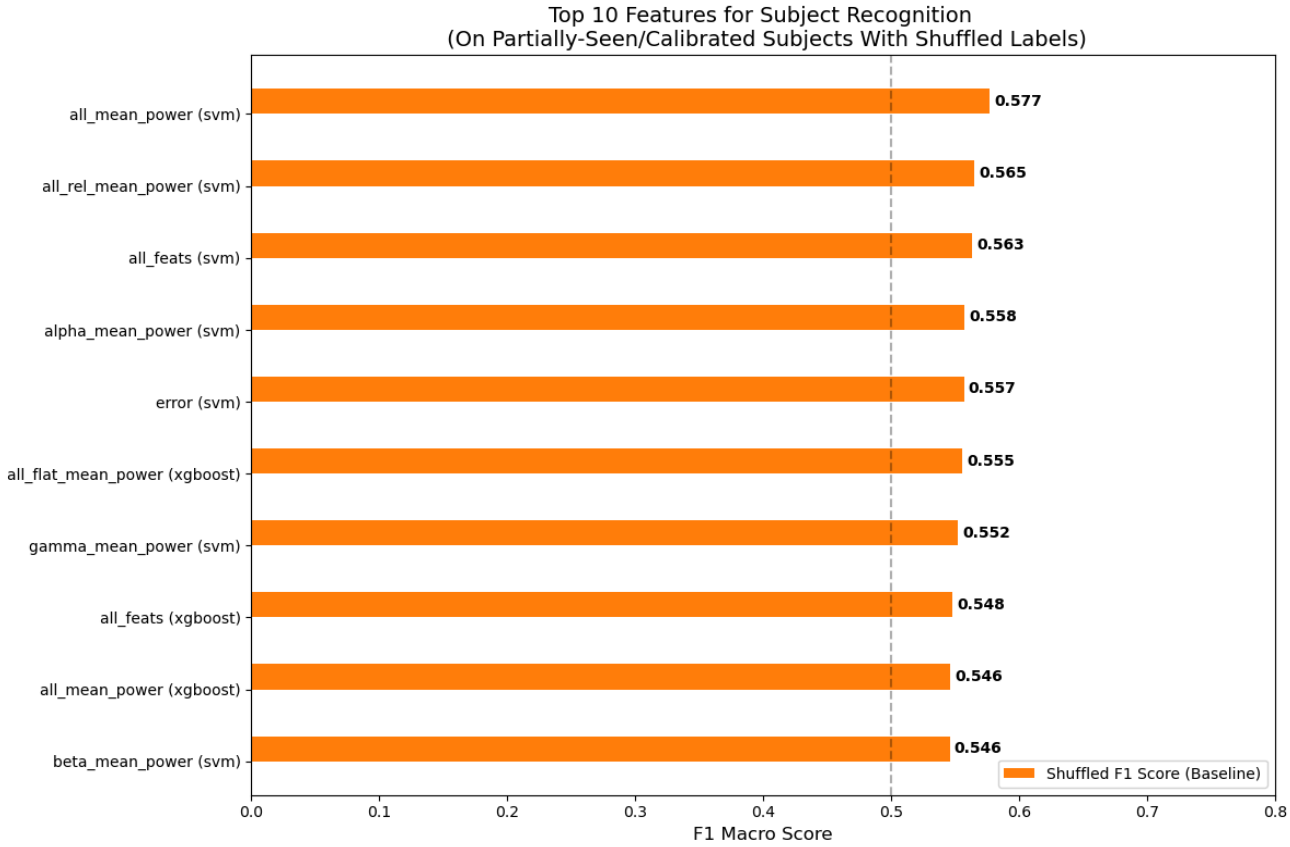

Figure 2: Top 10 feature-model combinations for subject recognition using shuffled labels. Absolute and relative mean power (SVM) were the strongest biometric identifiers, indicating that spectral features primarily capture individual identity.

Limitations

Subjectivity of Self-Reports and the “Ground Truth Problem”

The primary challenge in mind-wandering research is the reliance on subjective experience sampling. Because there is no objective behavioral marker for an internal thought process, the "ground truth" for the classifier relies entirely on the participant's self-assessment on a 0–3 scale. This introduces a significant introspective bias: a subject may be mind-wandering but, due to fatigue or to a lack of meta-awareness (Smallwood & Schooler, 2006), believe they are successfully meditating. Thankfully, this dataset contains subjects who are experienced meditators, meaning it is likely they have the meta-awareness to detect if they are meditating or mind-wandering. However, future applications of a neurofeedback system might not be able to depend on “calibrating” novice meditators with mind-wandering probes.

Data Quality

Training a robust classifier on EEG data is notoriously difficult due its low signal-to-noise ratio. Even after advanced preprocessing and ICA artifact rejection, EEG is easily contaminated with other non-brain sources like muscle tension or head movements. These artifacts can mask changes associated with mind-wandering and meditation.

The small sample size of 21 subjects limits the model’s ability to generalize to a broader population. There is likely a large amount of inter-subject variability as the way one person meditates can vary greatly from the way another meditates. Furthermore, only 762 datapoints

with a class imbalance makes it extremely difficult for a model to train on given how noisy EEG signal is.

Data Processing Constraints

1. Omitting Independent Component Analysis

Due to the scope of the project, Independent Component Analysis (ICA) was not implemented into the preprocessing pipeline. This likely added a large amount of noise to the EEG data and could also have added artifacts (eg. muscle tension or eye movement) that the model could have been training on instead of actual brain data.

2. Simplification of Temporal Data via Power Spectral Density

The feature engineering pipeline relied on collapsing 10-second EEG windows into stationary Power Spectral Density (PSD) features, resulting in a significant loss of temporal resolution. Given that mind-wandering is a transient and fluctuating phenomenon, averaging spectral power over such long epochs likely obscured the dynamic neural shifts occurring within each window. This simplification may have masked the subtler features that differentiate mind-wandering from meditative states.

3. Model Architectural Limitations

Lastly, Support Vector Machines (SVM) and XGBoost have limitations for EEG analysis. These classifiers are spatially agnostic, treating the 64-channel inputs as independent features and therefore overlooking the topographical relationships of electrode data. Furthermore, these models lack the sophistication (ex. compared to deep-learning) that can capture the complexity

of brain data. This lack of sophistication might explain why the models were unable to discern individual brain states yet recognize different subjects. Ultimately, without the capacity to model spatial and temporal hierarchies, these classifiers appear to have defaulted to identifying stable "neural fingerprints" rather than extracting the shared, state-dependent features necessary for generalized mind-wandering detection.

Future Directions

To overcome the challenges of overfitting and high inter-subject variability, future work will transition to use a larger dataset that was found shortly before the deadline of this project. This could provide the necessary statistical power necessary to learn these brain state features.

With a larger dataset, more sophisticated models could be used, such as Long Short-Term Memory networks (LSTM), Convolutional Neural Networks (CNNs), or Graph Neural Networks (GNNs). Using deep learning could allow for more nuanced representations of brain data, as well as spatial awareness if using CNNs and GNNs.

With more sophisticated models, raw time-series features could be used instead of simplified PSDs, giving the model more data to determine the subtle differences in brain-states. By leveraging more advanced models and richer features, one can move beyond simple subject-identification to a more robust brain-state detection model.

Distinction of Work

This project is structured to align with the goals of a Pattern Recognition course by focusing on a well-defined supervised learning pipeline. Although this work overlaps with my Honors Thesis

topic, the thesis will extend beyond this scope by exploring additional feature representations, alternative classifiers, and more in-depth statistical and neuroscientific analyses, culminating in a formal written thesis and defense. In contrast, this project is intentionally constrained to a focused approach using SVM and XGBoost that demonstrates the application of core pattern recognition principles to EEG-based mind-wandering detection.

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